

BONUS COMPANION GUIDE

140

DSAT Math Vocabulary Terms

The Words Behind Every Math Question on the Digital SAT

2026 EDITION

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A free companion to *The DSAT Vocabulary Advantage*

Welcome.

You bought a Reading and Writing vocabulary guide. But here's the secret most students miss: **the DSAT Math section is also a vocabulary test.**

Many wrong answers on Math come from the same hidden cause — a word or instruction you didn't fully understand. You knew the math. You just misread the question. "Approximate" vs. "exact." "At most" vs. "at least." "Linear" vs. "quadratic." "Median" vs. "mean." One word changes the entire problem.

This guide gives you the 140 most essential math vocabulary terms — the ones the Digital SAT uses again and again. Each term comes with a clean definition and a quick example. Study these, and you will stop losing easy points to misread questions. You will think faster. You will pick the right operation on the first try.

How to Use This Guide:

- Skim once to spot terms you're unsure about. Mark them.
- Master the four category sections. Each maps to a DSAT Math content area.
- Use the Quick Reference Cheat Sheet at the end for last-minute review.
- Pair this with the main vocabulary book to cover the whole test.

The Digital SAT Math Section at a Glance:

- 44 questions in 70 minutes (two adaptive modules)
- Calculator allowed throughout (built-in Desmos graphing calculator)
- Four content areas: Algebra, Advanced Math, Problem-Solving and Data Analysis, Geometry and Trigonometry
- Question types: multiple choice and student-produced response (grid-in)

Part 1 Algebra Essentials

35 terms covering linear equations, inequalities, systems, and functions

Variable. A letter representing an unknown number. *Example:* In $3x + 5$, x is the variable.

Coefficient. The number multiplied by a variable. *Example:* In $7x$, the coefficient is 7.

Constant. A fixed number with no variable. *Example:* In $2x + 9$, the constant is 9.

Term. A single number or variable, or numbers and variables multiplied together. *Example:* $3x$, -2 , and $5xy$ are each one term.

Expression. A combination of terms without an equals sign. *Example:* $4x + 7$ is an expression.

Equation. A statement that two expressions are equal. *Example:* $2x + 3 = 11$ is an equation.

Inequality. A statement comparing two expressions with $<$, $>$, $\leq$, or $\geq$. *Example:* $x + 5 > 10$ is an inequality.

Linear equation. An equation where the variable's highest power is 1; graphs as a straight line. *Example:* $y = 2x + 3$ is linear.

Slope. The steepness of a line; m in $y = mx + b$. Equals rise over run. *Example:* Slope of $y = 4x + 1$ is 4.

Y-intercept. The point where a line crosses the y-axis; b in $y = mx + b$. *Example:* Y-intercept of $y = 3x - 5$ is -5 .

X-intercept. The point where a graph crosses the x-axis (where $y = 0$). *Example:* X-intercept of $y = 2x - 6$ is $x = 3$.

Slope-intercept form. The equation $y = mx + b$, where m is slope and b is y-intercept. *Example:* $y = -2x + 7$.

Standard form. The equation $Ax + By = C$ for a line. *Example:* $3x + 4y = 12$.

Point-slope form. The equation $y - y_1 = m(x - x_1)$ using a known point and slope. *Example:* $y - 3 = 2(x - 1)$.

System of equations. Two or more equations solved together. *Example:* $\{y = 2x + 1, y = x + 4\}$.

Solution to a system. The point where the equations meet — the values that satisfy all equations. *Example:* $(3, 7)$ solves the system above.

Parallel lines. Lines with the same slope; they never intersect. *Example:* $y = 3x + 1$ and $y = 3x - 4$.

Perpendicular lines. Lines whose slopes are negative reciprocals; they meet at a right angle. *Example:* Slopes of 2 and $-\frac{1}{2}$.

Function. A rule that assigns exactly one output to each input. *Example:* $f(x) = 2x + 1$ is a function.

Domain. The set of all valid inputs for a function. *Example:* For $f(x) = \sqrt{x}$, domain is $x \geq 0$.

Range. The set of all possible outputs of a function. *Example:* For $f(x) = x^2$, range is $y \geq 0$.

Function notation. Writing $f(x)$ to show output value for input x . *Example:* If $f(x) = 3x$, then $f(4) = 12$.

Solve. Find the value(s) of the variable that make the equation true. *Example:* Solve $2x + 4 = 10$: $x = 3$.

Simplify. Rewrite an expression in its simplest form. *Example:* Simplify $3x + 2x = 5x$.

Substitute. Replace a variable with a given value or another expression. *Example:* Substitute $x = 2$ in $3x + 1$: gives 7.

Distribute. Multiply a term across each term in parentheses. *Example:* $3(x + 4) = 3x + 12$.

Like terms. Terms with the same variable raised to the same power. *Example:* $4x$ and $7x$ are like terms.

Absolute value. The non-negative distance of a number from zero, written $|x|$. *Example:* $|-5| = 5$.

Compound inequality. Two inequalities joined by “and” or “or.” *Example:* $-3 < x \leq 7$.

Direct variation. When $y = kx$ for some constant k ; Direct variation. When two quantities change proportionally, written $y = kx$ for some constant k . *Example:* $y = 5x$.

Inverse variation. When $y = \frac{k}{x}$; as one increases, the other decreases. *Example:* $y = \frac{12}{x}$.

Solution set. All values that satisfy an equation or inequality. *Example:* Solution set of $x > 3$ includes 4, 5, 6, ...

Consistent system. A system with at least one solution. *Example:* Lines that cross at one point.

Inconsistent system. A system with no solution; lines are parallel. *Example:* Same slope, different intercepts.

Equivalent equations. Equations with the same solutions. *Example:* $2x = 6$ and $x = 3$ are equivalent.

Part 2

Advanced Math

35 terms covering quadratics, polynomials, exponentials, and radicals

Quadratic equation. An equation where the highest power is 2. *Example:* $x^2 + 5x + 6 = 0$.

Parabola. The U-shaped graph of a quadratic function. *Example:* $y = x^2$ graphs a parabola.

Vertex. The maximum or minimum point of a quadratic graph. *Example:* Vertex of $y = x^2$ is $(0, 0)$.

Axis of symmetry. The vertical line through a parabola's vertex. *Example:* For $y = x^2 - 4$, it's $x = 0$.

Standard form (quadratic). $y = ax^2 + bx + c$. *Example:* $y = 2x^2 - 3x + 1$.

Vertex form. $y = a(x - h)^2 + k$, where (h, k) is the vertex. *Example:* $y = (x - 2)^2 + 3$.

Factored form. $y = a(x - p)(x - q)$, where p and q are the x-intercepts. *Example:* $y = (x - 1)(x - 5)$.

Quadratic formula. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, Quadratic formula. Solves any quadratic equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$. *Example:* Solves $ax^2 + bx + c = 0$.

Discriminant. $b^2 - 4ac$; tells the number of real solutions. *Example:* Positive: 2 solutions. Zero: 1. Positive: two real solutions. Zero: one real solution. Negative: no real solutions.

Factor. Rewrite an expression as a product of simpler terms. *Example:* $x^2 - 9 = (x - 3)(x + 3)$.

Roots / Zeros. The x-values where a function equals 0. *Example:* Roots of $y = x^2 - 4$ are $x = \pm 2$.

Polynomial. An expression with multiple terms involving whole-number exponents. *Example:* $3x^3 + 2x - 7$.

Degree. The highest power in a polynomial. *Example:* Degree of $5x^4 - x$ is 4.

Cubic function. A polynomial of degree 3. *Example:* $y = x^3 - 2x$.

Leading coefficient. The coefficient of the highest-power term. *Example:* In $-4x^3 + x$, leading coefficient is -4 .

Exponent. A number showing how many times to multiply a base by itself. *Example:* In 5^3 , the exponent is 3.

Base. The number being raised to a power. *Example:* In 5^3 , the base is 5.

Exponential function. A function with the variable in the exponent. *Example:* $y = 2^x$ is exponential.

Exponential growth. When a quantity multiplies by a constant rate over equal intervals. *Example:* Population doubles each year.

Exponential decay. When a quantity multiplies by a fraction over equal intervals. *Example:* Substance halves each year.

Growth factor. The multiplier in exponential growth, > 1 . *Example:* Doubling: growth factor = 2.

Decay factor. The multiplier in exponential decay, between 0 and 1. *Example:* Halving: decay factor = 0.5.

Radical. The root symbol $\sqrt{\quad}$ or any root expression. *Example:* $\sqrt{16} = 4$.

Radicand. The number under the radical. *Example:* In $\sqrt{25}$, radicand is 25.

Square root. A number that, multiplied by itself, gives the radicand. *Example:* $\sqrt{49} = 7$.

Cube root. A number that, multiplied by itself three times, gives the radicand. *Example:* $\sqrt[3]{27} = 3$.

Rational expression. A fraction containing polynomials. *Example:* $\frac{x+2}{x^2-1}$.

Rational equation. An equation containing rational expressions. *Example:* $\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$.

Extraneous solution. A solution that satisfies the modified equation but not the original. *Example:* Often appears after squaring or clearing fractions.

Composite function. A function applied to the output of another, written $f(g(x))$. *Example:* If $f(x) = x + 1$ and $g(x) = 2x$, $f(g(3)) = 7$.

Inverse function. A function that reverses another, written $f^{-1}(x)$. *Example:* Inverse of $f(x) = x + 4$ is $f^{-1}(x) = x - 4$.

Asymptote. A line a graph approaches but never touches. *Example:* $y = \frac{1}{x}$ has asymptotes at $x = 0$ and $y = 0$.

End behavior. What happens to a function's output as $x \rightarrow \pm\infty$. *Example:* For $y = x^2$, $y \rightarrow \infty$ on both ends.

Transformation. A change to a graph: shift, stretch, reflect. *Example:* $y = (x - 2)^2$ shifts $y = x^2$ right 2.

Imaginary number. A multiple of i , where $i = \sqrt{-1}$. *Example:* $3i$ is imaginary.

Part 3

Problem-Solving and Data Analysis

30 terms covering ratios, percentages, statistics, and probability

Ratio. A comparison of two quantities, often written $a : b$. *Example:* 3 : 4 means 3 parts to 4 parts.

Proportion. An equation showing two ratios are equal. *Example:* $\frac{2}{5} = \frac{4}{10}$.

Rate. A ratio comparing two different units. *Example:* 60 miles per hour.

Unit rate. A rate per single unit. *Example:* \$3 per pound.

Percent. A ratio out of 100. *Example:* 40 percent means 40 out of 100.

Percent change. $\frac{\text{new} - \text{old}}{\text{old}} \times 100$. *Example:* From 50 to 75 is a 50 percent increase.

Percent increase. A positive percent change. *Example:* Stock rose 20 percent.

Percent decrease. A negative percent change. *Example:* Price fell 15 percent.

Mean. The arithmetic average; sum divided by count. *Example:* Mean of 2, 4, 6 is 4.

Median. The middle value when data is ordered. *Example:* Median of 3, 5, 9 is 5.

Mode. The most frequent value in a data set. *Example:* Mode of 2, 2, 5, 7 is 2.

Range (statistics). The difference between maximum and minimum. *Example:* Range of 4, 7, 10 is 6.

Standard deviation. A measure of how spread out the data is. *Example:* Small SD means values cluster near the mean.

Outlier. A value far from the rest of the data. *Example:* In 5, 7, 8, 100 — the 100 is an outlier.

Distribution. The pattern of how data is spread. *Example:* Normal distributions are bell-shaped.

Skewed distribution. A distribution that leans left or right. *Example:* Skewed right means a long tail to the right.

Sample. A subset of a population used to make estimates. *Example:* 200 voters surveyed from a city.

Population. The full group being studied. *Example:* All voters in the city.

Random sample. A sample chosen so each member has an equal chance. *Example:* Reduces bias.

Margin of error. The range within which the true value is likely to fall. *Example:* ± 3 percentage points.

Confidence interval. A range likely to contain the true population value. *Example:* 95 percent confidence interval: 48 to 52 percent.

Correlation. A relationship between two variables. *Example:* Strong positive: both increase together.

Causation. One variable directly causes a change in another. *Example:* Correlation does not prove causation.

Scatterplot. A graph showing two variables as points. *Example:* X-axis: study hours; Y-axis: scores.

Line of best fit. A line that approximates the trend in a scatterplot. *Example:* Used to predict new values.

Probability. The likelihood of an event, between 0 and 1. *Example:* Flipping heads: probability 0.5.

Conditional probability. The probability of an event given another has occurred. *Example:* $P(\text{rain} \mid \text{cloudy}) = 0.6$.

Independent events. Events that do not affect each other. *Example:* Two separate coin flips.

Frequency. How often a value appears in data. *Example:* The number 7 appeared 4 times.

Two-way table. A table showing data classified by two variables. *Example:* Rows: gender. Columns: preference.

Part 4

Geometry and Trigonometry

40 terms covering lines, angles, triangles, circles, and trig functions

Point. A location with no size, marked by a dot. *Example:* Point A.

Line. A straight one-dimensional figure extending forever in both directions. *Example:* Line AB.

Line segment. Part of a line between two endpoints. *Example:* Segment $\overline{AB}$ has fixed length.

Ray. Part of a line with one endpoint extending forever in one direction. *Example:* Ray $\overrightarrow{AB}$.

Angle. Two rays sharing an endpoint, measured in degrees. *Example:* A 90-degree angle.

Acute angle. Less than 90 degrees. *Example:* A 45-degree angle is acute.

Right angle. Exactly 90 degrees. *Example:* Corner of a square.

Obtuse angle. Between 90 and 180 degrees. *Example:* A 120-degree angle.

Straight angle. Exactly 180 degrees. *Example:* A flat line.

Complementary angles. Two angles summing to 90 degrees. *Example:* 30 and 60 are complementary.

Supplementary angles. Two angles summing to 180 degrees. *Example:* 110 and 70 are supplementary.

Vertical angles. Opposite angles formed by two crossing lines; always equal. *Example:* X-shaped intersection.

Triangle. A three-sided polygon. *Example:* Interior angles sum to 180 degrees.

Equilateral triangle. All sides and all angles equal (each 60 degrees). *Example:* Perfect symmetry.

Isosceles triangle. Two sides equal; angles opposite those sides are equal. *Example:* Common on the DSAT.

Scalene triangle. All sides different lengths. *Example:* No symmetry.

Right triangle. Contains one 90-degree angle. *Example:* Satisfies the Pythagorean theorem.

Pythagorean theorem. In a right triangle, $a^2 + b^2 = c^2$, where c is the hypotenuse. *Example:* 3-4-5 triangle.

Hypotenuse. The longest side of a right triangle, opposite the right angle. *Example:* The diagonal side.

Similar figures. Same shape, different sizes; corresponding angles equal, sides proportional. *Example:* Scaled versions.

Congruent figures. Same shape and size. *Example:* Identical triangles.

Perimeter. Total distance around a shape. *Example:* Square with side 4: perimeter 16.

Area. The space inside a 2D shape. *Example:* Rectangle area: length times width.

Volume. The space inside a 3D shape. *Example:* Cube volume: side cubed.

Surface area. The total area of a 3D shape's outside. *Example:* Sum of all face areas.

Circle. All points equidistant from a center. *Example:* Perfectly round.

Radius. The distance from a circle's center to its edge. *Example:* Half the diameter.

Diameter. The distance across a circle through its center. *Example:* Twice the radius.

Circumference. The perimeter of a circle, equals $2\pi r$. *Example:* For $r = 3$: circumference 6π .

Area of circle. Equals πr^2 . *Example:* For $r = 4$: area 16π .

Arc. A portion of a circle's circumference. *Example:* A 90-degree arc covers one-quarter of the circle.

Central angle. An angle with its vertex at a circle's center. *Example:* Determines arc length.

Inscribed angle. An angle with vertex on the circle. *Example:* Equals half the central angle subtending the same arc.

Sine. In a right triangle, $\sin = \text{opposite} / \text{hypotenuse}$. *Example:* $\sin 30^\circ = 0.5$.

Cosine. In a right triangle, $\cos = \text{adjacent} / \text{hypotenuse}$. *Example:* $\cos 60^\circ = 0.5$.

Tangent. In a right triangle, $\tan = \text{opposite} / \text{adjacent}$. *Example:* $\tan 45^\circ = 1$.

Special right triangles. 30-60-90 and 45-45-90 triangles with predictable side ratios. *Example:* $1 : \sqrt{3} : 2$ and $1 : 1 : \sqrt{2}$.

Radian. A unit of angle measure; π radians equals 180 degrees. *Example:* $\frac{\pi}{2}$ radians is 90 degrees.

Unit circle. A circle of radius 1 used to define trig functions. *Example:* Coordinates give sine and cosine.

Coordinate plane. A 2D grid defined by x- and y-axes. *Example:* Point $(3, 4)$ is on it.

Cheat Sheet Last-Minute Review

The formulas and terms you must know cold

Essential Formulas

- **Slope:** $m = \frac{y_2 - y_1}{x_2 - x_1}$
- **Distance:** $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- **Midpoint:** $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
- **Quadratic formula:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- **Pythagorean theorem:** $a^2 + b^2 = c^2$
- **Area of triangle:** $A = \frac{1}{2}bh$
- **Area of circle:** $A = \pi r^2$
- **Circumference:** $C = 2\pi r$
- **Volume of rectangular prism:** $V = lwh$
- **Volume of cylinder:** $V = \pi r^2 h$
- **Percent change:** $\frac{\text{new} - \text{old}}{\text{old}} \times 100$
- **Exponential growth:** $y = a(1 + r)^t$
- **Exponential decay:** $y = a(1 - r)^t$

Words That Trip Students Up

- **At most** means $\leq$. **At least** means $\geq$.
- **Median** is the middle. **Mean** is the average. They are not the same.
- **Approximate** means rounded or estimated. **Exact** means no rounding.
- **Per** means divided by. "Miles per hour" = miles $\div$ hours.
- **Of** usually means multiply. "25 percent of 80" = 0.25×80 .
- **Difference** means subtract. **Sum** means add. **Product** means multiply. **Quotient** means divide.
- **Twice** means multiply by 2. **Half** means divide by 2.
- **Consecutive** means in order. Consecutive integers: 5, 6, 7.
- **Even** means divisible by 2. **Odd** is not.
- **Prime** means divisible only by 1 and itself.

Special Right Triangles

- **45-45-90:** sides in ratio $1 : 1 : \sqrt{2}$.
- **30-60-90:** sides in ratio $1 : \sqrt{3} : 2$ (opposite 30, 60, 90).
- **Common Pythagorean triples:** 3-4-5, 5-12-13, 8-15-17, 7-24-25.

You're Ready.

You now have 140 essential terms that appear across Digital SAT Math questions. When the test feels confusing, it is almost always vocabulary — not math — that creates the trap. Knowing the difference between “approximate” and “exact” will save you points. Knowing what “at most” means will save you points. Knowing a parabola has a vertex, an axis of symmetry, and roots will save you points.

The DSAT rewards students who read carefully and choose precisely. That is exactly the skill the main book trains. This bonus guide makes sure you carry the same skill into the Math section.

Good luck on test day. You earned this.

This is a free companion to *The DSAT Vocabulary Advantage*, 2026 Edition.

If this helped, please leave a review on Amazon. It helps other students find the book.

Updated for students preparing in 2026–27.